

Q.1 (a)

①

p : I run

q : I will get there quicker

P_1 : $p \Rightarrow q$

P_2 : q

$\therefore p$ Conclusion

The argument is invalid due to converse type of fallacy

(b) P_1 : $p \Rightarrow q$

P_2 : $\sim p$

$\therefore \sim q$

The argument is invalid due to inverse type of fallacy

(c) p : Bennett University is in Greater Noida.
 q : Bennett University is in UP.

P_1 : $p \Rightarrow q$

P_2 : q

p

The argument is invalid

(d) p : Jatin is a policeman
 q : Jatin is a footballer
 r : Jatin has big feet

$$\begin{array}{l}
 P_1: P \vee Q \\
 P_2: P \Rightarrow R \\
 P_3: \sim R \\
 \hline
 \therefore Q
 \end{array}$$

(2)

Answer

$$\begin{array}{l}
 \textcircled{1} P \vee Q \\
 \left\{ \begin{array}{l} \textcircled{2} P \Rightarrow R \\ \textcircled{3} \sim R \end{array} \right. \\
 \hline
 \textcircled{4} \sim P \\
 \textcircled{5} Q
 \end{array}$$

By Modus Tollens with $\textcircled{2}$ & $\textcircled{3}$

By disjunctive Syllogism in $\textcircled{1}$ & $\textcircled{4}$

So the argument is valid.

(But not sound as the 2nd premise is not true in real life)

(2)

P: The Company invests in renewable energy

Q: The Company will reduce its carbon footprint.

R: The Company faces public backlash

$P_1: P \Rightarrow Q$

$P_2: P$

$P_3: Q \vee R$

Conclusion $\sim R$

$\textcircled{1} P \Rightarrow Q$ (given)

$\textcircled{2} P$ (given)

$\textcircled{3} Q \vee R$ (given)

$\textcircled{4} Q$ By using $\textcircled{1}$ & $\textcircled{2}$ (M.P.)

$\textcircled{5} \sim R \times$

The argument is invalid due to the

fallacy of the type disjunctive syllogism.

(f)

2

P: A student attend lectures of DMS

Q: Student's understanding of CS will improve.

R: Student is not present on campus

S: DMS examination is not tough

Given ① $P \Rightarrow Q$ (given)

② $\sim Q$ (given)

③ $\sim P \Rightarrow R \vee S$ (given)

④ $\sim R$ (given)

⑤ $\sim P$ By MT in ① & ②

⑥ $R \vee S$ By MP in ③ & ⑤

⑦ S By disjunctive syllogism in ④ & ⑥

To Prove
Conclusion S

The argument is valid

Tutorial - 3 Solution

④

Q.2 (a)

$A(x)$: x is a lawyer

$B(x)$: x is a liar

$$\therefore \forall x (A(x) \Rightarrow B(x))$$

(b) $\exists x (A(x) \wedge B(x))$

(c) $P(x)$: x is a person

$Q(x)$: x becomes politician

$R(x)$: x will get corrupt

$$\therefore \forall x (P(x) \wedge Q(x) \Rightarrow R(x))$$

(d) $P(x)$: x is a doctor

$Q(x)$: x know Calculus

$$\therefore \forall x (P(x) \Rightarrow \neg Q(x))$$

or $\neg \exists x (P(x) \wedge Q(x))$

(e) $P(y, x)$: y is greater than x

$Q(x)$: x is a positive integer

$$\forall x (Q(x) \Rightarrow \exists y (Q(y) \wedge P(y, x)))$$

(f) $P(x)$: x is an integer

$Q(x)$: x is positive

$R(x)$: x is negative

$$\forall x (P(x) \Rightarrow (Q(x) \vee R(x)))$$

(5) (2)

Q.3 (a) Let UOD = All Indian Festivals
universe of discourse

$P(x)$: x is celebrated based on solar calendar
 $Q(x)$: x is celebrated based on lunar calendar

$$\therefore \forall x (P(x) \vee Q(x))$$

II But Let UOD = All Festivals

$P(x)$: x is an Indian Festival

$Q(x)$: x is celebrated based on solar calendar

$R(x)$: x is celebrated based on lunar calendar

$$\therefore \forall x (P(x) \Rightarrow (Q(x) \vee R(x)))$$

(b) Let UOD = universe of citizens

$P(x)$: x is eligible to vote

$Q(x)$: x is above 18 years of age

$$\forall x (P(x) \Leftrightarrow Q(x))$$

(c) Let UOD = universe of family members

$P(x)$: x enjoys spicy food

$Q(x)$: x dislikes sweet

$$\forall x ((P(x) \wedge Q(x)) \vee (\neg Q(x) \wedge P(x)))$$

$$(d) \quad \forall x (Even(x) \wedge x > z) \Rightarrow (\exists p \exists q: Prime(p) \wedge Prime(q) \wedge (x = p + q))$$

(6)

Q. 4 (a) There is only one book on the table
or
Exactly one book is on the table

(b) Every prime number greater than 2 is odd.

(c) There exists a real number x such that, when added to any real number y , the result ~~is~~ is y .
or

There is atleast one real number x , for all real numbers, such that addition of x in any real number, the result will be the same real number

NOTE: Defining Zero Element i.e. $x=0$

Q. 5 (a) True { for $x=9$ $R(9)$ is true }

(b) False { for $y=10$ }

(c) True { for any real number, there always exist one real number that is less than it. }

(d) False { There does not exist any real number which is greater than or equal to all other real numbers }

(e) True { for any two real numbers x & y , either $x > y$ or $x < y$ }

(f) True { As there exist at least one real number greater than 9 and all real numbers are not equal to 9 }

(g) False { There does not exist any real number greater than 3 which is greater ~~than~~ than or equal to all real numbers }

(h) True { As $\forall x: R(x)$ is false

$\Rightarrow \forall x \forall y: R(x) \wedge S(y)$ is false

$\Rightarrow \forall x \forall y: (R(x) \wedge S(y)) \Rightarrow Q(x, y)$ is true

Q.6 (a) dom = all foods

$H(x)$: x is healthy to eat

$G(x)$: x tastes good

$Y(x)$: You eat x

$\forall x (H(x) \rightarrow \neg G(x))$ — (1)

$H(\text{tofu})$ — (2)

$\forall x (Y(x) \rightarrow G(x))$ — (3)

$\neg Y(\text{tofu})$

To check its validity

apply universal instantiation in (1) and get

$H(\text{tofu}) \rightarrow \neg G(\text{tofu})$ — (4)

apply modus ponens in (4) & (2) and we get

$\neg G(\text{tofu})$ — (5)

apply universal instantiation in (3) and get

$Y(\text{tofu}) \rightarrow G(\text{tofu})$ — (6)

apply modus tollens in (5) & (6) and get

$\neg Y(\text{tofu})$

which is our conclusion. So it is a valid.

(b)

domain = all people

$$\forall x (B(x) \Rightarrow I(x))$$

$$\exists x (B(x) \wedge \neg S(x))$$

$$\forall x (I(x) \Rightarrow \neg S(x))$$

(8)

$B(x)$: x is from Bennett University

$I(x)$: x lives within 50 km of Indira Gate

$S(x)$: x has seen the Indira Gate

It is not valid because the conclusion can never be true for all elements if one of the premise is existentially true. In above question, premise 2 is existentially true.

(c) domain = all particles

$P(x)$: x is proton

$N(x)$: x is neutron

$S(x)$: x is subatomic particle

$$\forall x (P(x) \Rightarrow S(x))$$

$$\forall x (N(x) \Rightarrow S(x))$$

$$\forall x (P(x) \Rightarrow N(x))$$

It is not valid. It is a fallacy of Hypothetical Syllogism

Q7

9

(a) Every student is honest

Predicate logic:

~~$P(x)$~~ : x is a student

$Q(x)$: x is honest

$$\forall (x) (P(x) \Rightarrow Q(x))$$

Negation logic:

$$\exists x (P(x) \wedge \neg Q(x))$$

There exists at least one student who is not honest

(b) There is a teacher who teaches Mathematics

Predicate logic:

$P(x)$: x is a Teacher

$Q(x)$: x teaches mathematics

$$\exists x (P(x) \wedge Q(x))$$

Negation logic:

$$\forall x (P(x) \Rightarrow \neg Q(x))$$

No teacher teaches mathematics

(c) Every Employee reports to some manager

Predicate logic:

$P(x)$: x is an employee

$Q(y)$: y is a manager

$R(x, y)$: x reports to y

$$\forall x (P(x) \Rightarrow \exists y (Q(y) \wedge R(x, y)))$$

Negation logic:

There exists an employee who does not report to any manager

(d) There is a person who likes everyone

Predicates :-

$$P(x, y) : x \text{ likes } y$$

$$\exists x \forall y P(x, y)$$

Negation logic:-

$$\forall x \exists y \neg P(x, y)$$

Everyone dislikes at least one person

(e) Everyone has a friend

Predicate :-

$$P(x, y) : x \text{ is a friend of } y$$

~~$$\forall x \exists y P(x, y)$$~~

$$\forall x \exists y P(x, y)$$

Negation:-

$$\exists x \forall y \neg P(x, y)$$

There exists someone who has a friend.